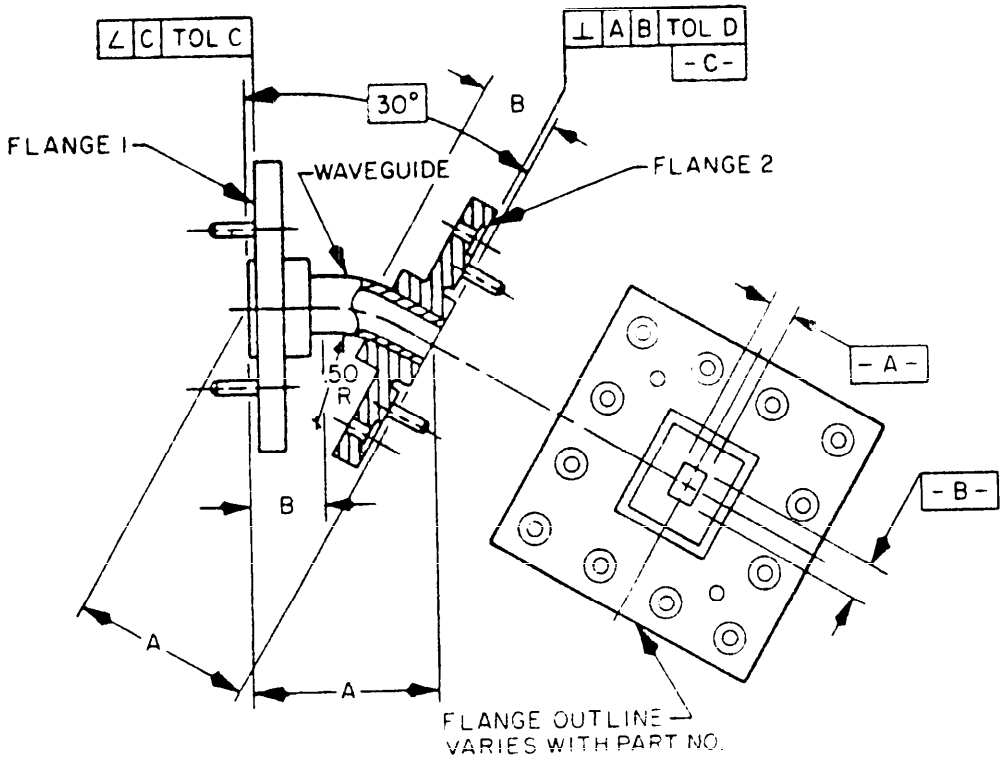
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, 30-DEGREE H-PLANE BEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

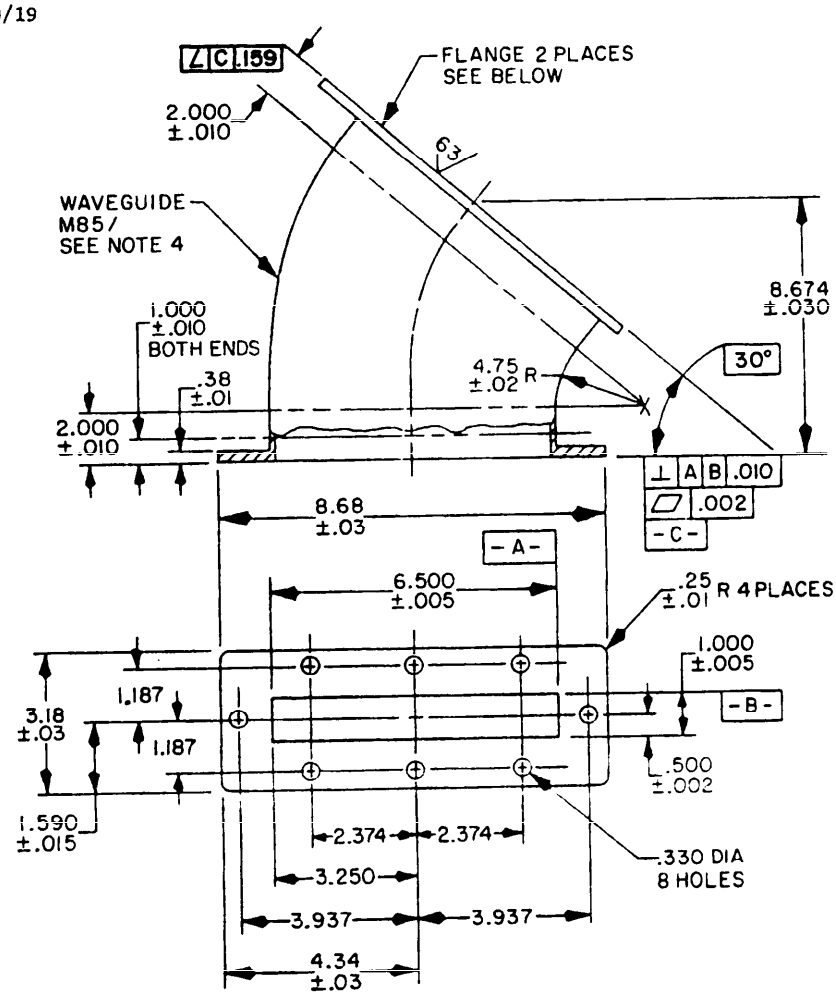


Configuration A
Part numbers M3970/19-002 thru -007.

FIGURE 1. Waveguide assembly.

FSC 5985

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Configuration B
Part number M3970/19-001.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.030	.76	.500	12.70	2.374	60.30	4.75	120.65
.005	.13	.159	4.04	1.000	25.40	3.18	80.77	6.500	165.10
.010	.25	.25	6.35	1.187	30.15	3.250	82.55	8.674	220.32
.015	.38	.330	8.38	1.590	40.39	3.937	100.00	8.68	220.47
.020	.51	.38	9.65	2.000	50.80	4.34	110.24		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.002 (.05 mm).

FIGURE 1. Waveguide assembly - Continued.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions.

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
001	Configuration B	Configuration B		1.25 - 1.35		1.05	5	7.732 (196.39) ±.030 (.76)	2.00 (50.80) ±.010 (.25)	.175 (4.45)	.004 (.10)	4.75 (120.65) ±.030 (.76)
002	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	30	1.230 (31.24) ±.015 (.38)	0.500 (12.70)	.019 (.48)	.001 (.03)	.5 (12.70) ±.015 (.38)
003	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	---	↓	↓	.020 (.51)	.001 (.03)	↓
004	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	---	↓	↓	.020 (.51)	.001 (.03)	↓
005	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	---	↓	↓	.019 (.48)	.001 (.03)	↓
006	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	---	↓	↓	.018 (.46)	.001 (.03)	↓
007	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	---	↓	↓	.018 (.46)	.001 (.03)	↓

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ Waveguide per M85/1-018 except B dimension shall be 1.000 and D dimension 1.500.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Power handling capability for dash no. 001 only: 350 kW peak, 2.2 kW average, at a frequency of 1.35 GHz, 50°C temperature.

Part number: M3970/19-(dash number from table I).

NOTE: Assemblies covered by this specification are substitutable for the following commercial types:

Part number M3970/19-	Manufacturer's code	Commercial type
001	49956	712066
002	73293	44321H-302
003	73293	44322H-302
004	73293	44323H-302
005	73293	44324H-302
006	73293	44325H-302
007	73293	44326H-302

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - MU
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0835-16)

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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